How Many Categories Are Enough? Distribution-Free Certification Limits for Few-Shot Anomaly Thresholds

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ABSTRACT

At the nominal false-alarm level $\alpha = 0.20$, a target-only leave-one-image-out (LOIO) alarm reaches an empirical false-alarm rate of 0.341 on Gaussian-corrupted MVTec at $k = 4$, approximately 1.7 times the nominal level. This motivates a study-design question: how much source evidence, measured in images or independent categories, can support threshold certification for a new category? Using a frozen DINOv2 principal component analysis (PCA) residual ranker, we audit 15 MVTec and 12 VisA categories under four corruptions. Target-only p-values have the finite resolution floor $1/(k+1)$, yet crossing this floor does not restore false-alarm stability under shift. We derive a category-count feasibility calculus. Even with zero observed category losses, an optimistic multiplicity-free, deterministic, uniformly valid, distribution-free 95% upper bound requires at least 14, 29, and 59 independent categories at $\alpha = 0.20$, 0.10, and 0.05; simultaneous frozen-protocol requirements are higher. Cross-category Reliability Estimation with Source Support (CRESS) operationalizes this audit by separating reference, threshold-proposal, and certification categories. With only three or four certification categories, every category-certified target cell across 960 frozen gate configurations receives the fail-closed threshold $\tau^\star = 0$; the smallest category-level upper bound is 0.950, far above the largest tested α. Under an independent and identically distributed (iid) source-image model, image-unit sensitivity bounds select positive thresholds in 36.7% to 60.3% of target cells, but address the selected source mixture rather than marginal risk over a new-category draw. The main contribution is a quantitative feasibility boundary and an estimand-aware protocol for deciding when source evidence can, and cannot, justify a transferable reliability claim.

1. Introduction

At the nominal false-alarm level $\alpha = 0.20$, the target-only alarm studied in this paper reaches an empirical false-alarm rate (FAR) of 0.341 on Gaussian-corrupted MVTec at $k = 4$. This is the largest dataset-level aggregate in the frozen corruption grid and is approximately 1.7 times the requested level. It is not a population-level failure probability: it is a benchmark observation pooled over categories and five nested support seeds under one declared synthetic shift. Nevertheless, it exposes an operational gap that ranking metrics alone cannot reveal. A detector may assign defects higher anomaly scores than normal images and obtain a strong area under the receiver operating characteristic curve (AUROC) or average precision (AP), while a fixed score threshold still produces substantially different false-alarm behavior across categories or input conditions.

This gap is particularly consequential in few-shot industrial anomaly detection (AD). A target category commonly provides only $k \in \{1, 2, 4, 8\}$ known-normal support images, whereas anomalous examples are rare, heterogeneous, and unavailable for threshold fitting. Frozen visual foundation models make ranking feasible in this regime: DINOv2 [1] patch features support nearest-neighbor memory banks, as in PatchCore-style methods and AnomalyDINO [2, 3], or compact PCA normal subspaces, as in SubspaceAD [4]. These advances do not by themselves specify which alarm threshold has a stable operational meaning.

Two responses are natural. The first is to calibrate on the k target-normal supports. Our target-only route evaluates each support image in a LOIO fit and ranks a test score against the resulting normal scores. This construction is label-free and interpretable, but its p-values occupy only the grid $\{j/(k+1)\}$. Increasing k refines that grid; it does not ensure that support and shifted test scores satisfy the exchangeability conditions needed for false-alarm validity. The empirical FAR of 0.341 is an example of this distinction: the operating level is attainable, yet the shifted normal p-values are anti-conservative.

The second response is to borrow normal evidence from non-target source categories. A larger source archive offers finer empirical ranks and can propose thresholds below the target-only grid. It also changes the statistical question. Independent image draws from a declared source mixture can estimate risk for that mixture, whereas a guarantee intended for a previously unseen category depends on evidence across independent categories. Repeated images, corruption views, or support seeds do not create new category units. The central question of this paper is therefore:

> How much source evidence, measured in which statistical unit, is needed to certify an anomaly threshold for a new category?

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We answer this question at three levels. First, for any deterministic, uniformly valid, distribution-free 95% upper confidence bound based only on iid category losses, an all-zero sample cannot certify $\alpha = 0.20, 0.10$, or 0.05 with fewer than 14, 29, or 59 categories, respectively, even without multiplicity. These are optimistic impossibility-side limits, not sufficient sample sizes for the implemented procedure. Under the frozen simultaneous allocation, the corresponding lower limits are at least 22, 46, and 94 for one candidate, while the declared Hoeffding rule requires 60, 240, and 958. Second, we audit target-only normal p-values under four controlled corruptions and show that crossing the finite resolution floor does not imply stable FAR. Third, we use CRESS to separate source categories into disjoint reference, threshold-proposal, and certification roles and test whether the available archive reaches the predicted feasibility boundary.

CRESS is a source-assisted thresholding and audit protocol, not a new anomaly ranker. The inherited ranking substrate is a frozen DINOv2 PCA residual. The reference categories define a source-reference rank map, the proposal categories define a finite candidate set, and the certification categories evaluate fixed candidates with a family-adjusted upper confidence bound (UCB). The protocol returns the largest passing threshold or the fail-closed value zero. Source-risk dominance can transfer the bound to a particular target category. Category exchangeability supports only a marginal bound averaged over a new-category draw, not category-conditional control for every realized target. Normalization and source pooling establish neither condition.

The strict audit contains only three certification categories in VisA cells and four in MVTec and transfer cells. Accordingly, every category-certified target cell receives $\tau^\star = 0$ across the 960 frozen configurations. This is not 960 independent failures, nor a compute or optimization failure. The configurations span jobs, source-view modes, support sizes, nominal levels, and conditions, but share the same category-count obstruction. Their value is to confirm that the analytically predicted boundary binds throughout the declared grid. By contrast, under the declared iid source-image model, image-unit analyses have 72 to 191 units per cell and select positive thresholds in 36.7% to 60.3% of target cells, depending on the source-to-target job. This contrast is not a stronger-versus-weaker version of one certificate: the two analyses target different risks. The same source archive is statistically large when counted in images but insufficient when the deployment claim concerns marginal risk for a new-category draw.

The paper makes three contributions:

- **A category-count feasibility calculus.** We derive the Hoeffding-specific requirement and a broader all-zero lower bound for deterministic uniformly valid distribution-free category-risk upper bounds. The resulting calculations quantify the minimum budget of independent categories implied by the requested risk and confidence levels and serve as a study-design tool rather than a post hoc explanation alone.
- **An operational shift audit.** Across 15 MVTec and 12 VisA categories, the target-only analysis distinguishes unattainable nominal levels from attainable but unstable operating points. The largest dataset-level aggregate FAR is 0.341 on Gaussian-corrupted MVTec at $k = 4$ and $\alpha = 0.20$; VisA is conservative in the corresponding cells, showing that the shift effect is dataset-dependent rather than universal.
- **An estimand-aware source protocol.** CRESS separates reference, proposal, and certification categories, retains zero fallbacks, and reports image-mixture and category-level conclusions without substituting one for the other. The frozen strict audit verifies the predicted category shortage within datasets and for transfer from MVTec to VisA and the Metal Parts Defect Dataset (MPDD); direct pooling and compact-ranker results are reported only as supporting empirical context.

Figure 1 previews the central design calculation. The black curve is the optimistic multiplicity-free distribution-free floor; the blue and orange curves add the frozen family allocation and the declared Hoeffding rule in their most favorable one-candidate case. A positive certificate is infeasible whenever the relevant curve remains above the requested α. The shaded region marks the three or four certification categories available in the strict audit; Table 1 later reports the exact candidate-family calculations.

The remainder of the paper is organized as follows. Section 2 positions the study against few-shot detectors, calibration, and conformal risk control. Section 3 defines the inherited scorer, target-only route, source construction, CRESS protocol, and certificate scope. Section 4 specifies the frozen datasets, statistical units, partitions, and claim rules. Section 5 proceeds from the shift audit to category-count feasibility and strict CRESS, followed by the supporting ranking and storage analysis. Sections 6 and 7 state the boundaries and implications.

2. Related Work

2.1. Few-Shot Industrial Anomaly Detection

Classical industrial AD typically assumed access to normal-only training data and identified deviations at test time. Patch distribution, self-supervised, and reconstruction-discriminative approaches such as PaDiM, CutPaste, and DRAEM provided important context for this literature [5–7]. PatchCore [2] established memory-bank patch retrieval as a robust baseline by storing representative normal features and scoring test patches using nearest-neighbor distance. AnomalyDINO [3] extended this paradigm to few-shot inspection with frozen DINOv2 representations. Subsequent work introduced registration-based adaptation (RegAD [8]), graph-structured memory (GraphCore [9]), fast feature reconstruction (FastRecon [10]), vision-language zero- and

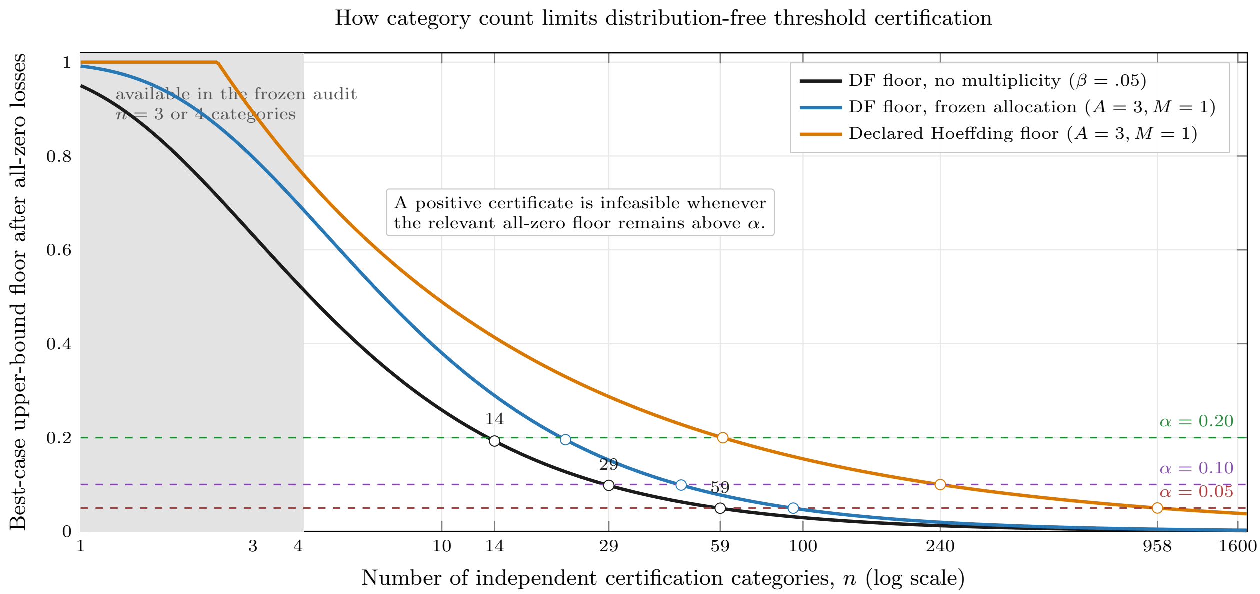

Figure 1: Analytic category-count feasibility after observing zero loss in every certification category. The black curve is the multiplicity-free distribution-free lower bound $1 - 0.05^{1/n}$. The blue curve applies the frozen allocation $\beta = \delta/(2AM)$ with $\delta = 0.05$, $A = 3$, and the favorable case $M = 1$; the orange curve is the corresponding Hoeffding floor. Horizontal lines mark the requested risk levels, and the shaded region marks the three or four independent certification categories available in the frozen audit. The labels 14, 29, and 59 are optimistic necessary counts without multiplicity, not sufficient sample sizes for CRESS. Curves above a requested α indicate that a positive threshold cannot be certified even under all-zero observed category loss.

few-shot scoring (WinCLIP [11], CLIP-SAM collaboration [12], and cross-modal prompt regularization [13]), and in-context residual learning (InCTRL [14]). These methods were effective rankers, but their reference state could remain nontrivial at inference time and the raw score scale did not represent a calibrated probability. Calibration under distribution shift was also known to degrade for deep models generally [15]. Our work is complementary: it does not compete on ranking but instead examines how few-shot detectors should expose decision reliability.

2.2. Frozen-Feature Subspace Detectors

Subspace approaches replaced explicit patch retrieval with a compact normal subspace. SubspaceAD [4] demonstrated that frozen DINOv2 patch features and PCA or subspace residuals were already effective for few-shot anomaly ranking, establishing the ranking mechanism as prior art. Our contribution begins after the residual is computed and examines the reliability claims that a low-storage subspace detector can support under few-shot and distribution-shift conditions.

2.3. Calibration and Decision Reliability in Few-Shot AD

Calibration measured agreement between predicted confidence and empirical correctness [16, 17]. Standard post-hoc methods included Platt scaling and temperature scaling [16, 17]. Histogram binning and isotonic regression provided nonparametric alternatives [18, 19]. Sparse labels, open-ended anomaly types, and category-dependent scores made this objective especially difficult in anomaly detection. Prior reliability analyses of few-shot AD based on DINOv2 also documented calibration and perturbation sensitivity [20]. These findings motivated our separation of ranking quality from an operational question: whether a declared alarm rule maintains its requested false-alarm behavior under a specified shift. A calibrated probability summary alone would not constitute such a risk guarantee.

2.4. Conformal Prediction and Risk-Controlled Calibration

Conformal prediction transformed nonconformity scores into p-values under exchangeability assumptions [21–23]. This approach was particularly suitable for normal-only few-shot AD because normal support images could define a reference distribution without requiring real anomalies. Conformal anomaly detection was already well established: leave-one-out, bootstrap, and cross-conformal anomaly detectors had been directly analyzed [24]; weighted conformal detection in low-data regimes had examined resolution and effective-sample-size effects [25]; general nonconformity toolkits had been developed [26]; and few-shot conformal prediction with auxiliary tasks had addressed small target calibration sets [27]. Conformal AD, LOIO calibration, auxiliary-task transfer, and weighted conformal p-values were therefore established mechanisms.

Risk-controlling calibration also preceded our work. Learn then Test cast finite-sample calibration as a multiple-testing problem over candidate configurations [28], while conformal risk control selected a parameter to control the expectation of a monotone loss [29]. In anomaly detection, PAC-Wrap provided semi-supervised probably approximately correct (PAC) bounds for false-positive and false-negative rates around a base detector [30]. These works established the general principles of candidate testing, held-out risk assessment, and one-sided error bounds; we do not claim those statistical primitives as new. CRESS instead instantiates them for normal-only cross-category thresholding by assigning disjoint source categories to reference, proposal, and certification roles, distinguishing selected-mixture image risk from marginal new-category risk, and auditing whether the available number of independent categories can support a nonzero threshold.

Structure-aware conformal adaptation had also been studied outside anomaly detection. Lam and Anh proposed HeAD-CP, which adapted graph-score diffusion using label-free local-homophily estimates and showed that uniform diffusion could impair prediction-set efficiency on heterophilic graphs while preserving marginal coverage [31]. This result supported the broader need to audit reliability transformations under structural shift, but its setting concerned graph neural network (GNN) node-classification sets rather than few-shot industrial anomaly alarms or cross-category source certification.

The deterministic source-view choices were conceptually informed by the shared-versus-specialized selection motif in Shape-Adapting Gated Experts (SAGE) [32]. That work concerned learned expert selection rather than certification of anomaly thresholds. CRESS uses no learned expert gate: its source-view modes are predeclared protocol choices, and its reference, proposal, and certification partitions have disjoint statistical roles. The citation therefore records design inspiration without attributing CRESS validity or novelty to the SAGE architecture.

The present contribution begins at the independent-category requirement of a simultaneous source certificate, with the target-only grid $1/(k+1)$ serving as a motivating finite-resolution constraint. This work establishes the Hoeffding-specific and distribution-free all-zero category-feasibility bounds, empirically audits corruption-shift deviations, and evaluates source-assisted candidate thresholds without target anomaly labels. CRESS does not introduce a new nonconformity score or generic confidence bound; it specifies how source categories are separated into reference, proposal, and certification roles and audits whether the resulting source certificate can be nonvacuous at the available category count.

3. Method

3.1. Problem Setup

For each target category c, we observe a support set $\mathcal{S}_c = \{x_i\}_{i=1}^{k}$ containing only normal images. Target test images may be normal or anomalous, but their labels are reserved exclusively for evaluation and do not enter detector fitting, threshold proposal, or certification. Other categories may additionally provide a declared source-normal archive. The framework produces a fixed raw anomaly score $s_c(x)$ and optional pixel map, a target-only reliability value $p_{\mathrm{LOIO}}(x)$ when leave-one-image-out calibration is available, and, for the source-assisted route, a threshold $\tau^\star$ accompanied by a source-domain certificate. Figures 2 and 3 separate the inherited ranking substrate, the target-only route, and the source-assisted route.

The statistical unit is fixed before threshold evaluation. Under its iid sampling model, an image-unit analysis targets the mean alarm loss for a draw from the declared source-image mixture. A category-unit analysis first averages alarms within each source category and then targets the mean loss of a new draw from the declared source-category meta-population. The latter requires independent category units; additional images, support seeds, or corruption views within an existing category do not increase their number. Neither source estimand is automatically the target-category risk. That transfer requires the additional condition stated at the end of this section.

3.2. Frozen Patch Features and Subspace Ranking

We extract patch features with a frozen DINOv2 small Vision Transformer using 14×14 patches (ViT-S/14). Let $z_{ij} \in \mathbb{R}^d$ denote patch j from image i. A PCA subspace is fit on the target support patches. With support mean μ_c and retained principal directions $U_c \in \mathbb{R}^{d\times m}$, the patch residual is

$$r_c(z) = \left\|(z-\mu_c) - U_c U_c^\top (z-\mu_c)\right\|_2 . \tag{1}$$

The patch residuals form an anomaly map. The image-level ranking score is

$$s_c(x) = \mathrm{Agg}_j \; r_c(z_j). \tag{2}$$

The clean accuracy-storage benchmark uses the patch maximum, whereas the conformal protocols use the mean of the top-1% patch residuals ($\rho = 0.01$) as a smoothed maximum. AUROC and AP are always computed from the fixed score $s_c(x)$ within a declared protocol; none of the subsequent reliability transformations changes that score or its ranking.

3.3. Target-Only LOIO Reliability

For $k \geq 2$, the target-only route constructs normal calibration residuals by leave-one-image-out support splitting. For each support image x_i, a subspace is fit on $\mathcal{S}_c \setminus \{x_i\}$, and the held-out image receives the score $s_{c,-i}(x_i)$. The resulting set

$$\mathcal{R}_{c,\mathrm{LOIO}} = \{s_{c,-i}(x_i) : x_i \in \mathcal{S}_c\} \tag{3}$$

serves as target-normal nonconformity evidence. A test image, scored using the subspace fit on all k supports, receives

$$p_{c,\mathrm{LOIO}}(x) = \frac{1 + \sum_{r\in\mathcal{R}_{c,\mathrm{LOIO}}} \mathbf{1}\{r \geq s_c(x)\}}{1 + |\mathcal{R}_{c,\mathrm{LOIO}}|}. \tag{4}$$

We report $1 - p_{c,\mathrm{LOIO}}(x)$ only as a probability-like anomaly reliability view. It measures extremeness relative to held-out support normals and is not a supervised anomaly posterior.

Placeholder: target-side scoring and reliability pipeline

Panel (a): frozen DINOv2 features → support-fitted PCA residual scorer → image score and pixel anomaly map.

Panel (b): LOIO support scores → target-only rank transform $p_{c,\mathrm{LOIO}}(x)$ → alarm at $p_{c,\mathrm{LOIO}}(x) \leq \alpha$, with resolution floor $1/(k+1)$.

Figure 2: Target-side scoring and reliability routes. Panel (a) uses frozen DINOv2 patch features and a support-fitted PCA residual scorer to produce the fixed image-ranking score and pixel anomaly map; performance metrics are evaluation-only. Panel (b) reuses the scoring construction in leave-one-image-out folds to form target-normal calibration scores and the target-only rank value $p_{c,\mathrm{LOIO}}(x)$. The smallest attainable value is $1/(k+1)$, and the alarm rule does not alter the underlying ranking score. The final artwork will replace this placeholder without changing the method.

The construction is asymmetric: each calibration score uses $k-1$ support images, whereas the test score uses all k. Equation (4) is therefore a LOIO approximation in the spirit of cross-conformal inference [21, 24, 33], not a split-conformal p-value with an automatic finite-sample guarantee [34, 35]. A matched-LOIO audit scores held-out support and test images under the same fold-specific subspaces, but does not improve the observed false-alarm and power trade-off. We consequently retain the simpler construction and audit its normal p-values empirically rather than assuming exchangeability.

3.4. Target-Only Resolution

Remark 1 (attainable-alpha floor). For any finite calibration scores $r_1, \ldots, r_k$, a p-value of the form in Eq. (4) belongs to $\{j/(k+1) : j = 1, \ldots, k+1\}$. Hence the alarm $\mathbf{1}\{p(x) \leq \alpha\}$ is identically zero for $\alpha < 1/(k+1)$.

Proof. The exceedance count is an integer from 0 to k. Adding one and dividing by $k+1$ gives the stated grid and lower bound. No exchangeability assumption is needed for this algebraic claim. □

For $k = 4$, the floor is 0.2; for $k = 8$, it is $1/9 \approx 0.111$. The attainable-alpha analysis reports, for each nominal α, the largest attainable grid point not exceeding α and the corresponding empirical false-alarm rate. The floor is a finite-resolution property, not an algorithmic innovation or evidence that the underlying ranker cannot separate anomalies. Standard randomization can smooth this deterministic grid under the relevant exchangeability assumptions, but it cannot by itself repair support-to-test shift.

3.5. Source Score Construction and View Selection

CRESS uses known-normal images from categories other than the target. For every target or source category c, let $\mathcal{B}_c$ denote its support-only calibration residuals. For $k \geq 2$, $\mathcal{B}_c = \mathcal{R}_{c,\mathrm{LOIO}}$. Because LOIO is undefined with one support image, the declared $k = 1$ stress test uses one patch-split support calibration residual instead. We summarize $\mathcal{B}_c$ by its median and median absolute deviation (MAD):

$$\begin{aligned} m_c &= \mathrm{median}(\mathcal{B}_c), \\ d_c &= \mathrm{MAD}(\mathcal{B}_c), \\ \widetilde{s}_c(x) &= \frac{s_c(x) - m_c}{\max(d_c, \varepsilon)}. \end{aligned} \tag{5}$$

Here $\varepsilon = 10^{-6}$ is fixed before evaluation. This support-only normalization makes cross-category score scales more comparable; it does not establish equality or exchangeability of their distributions. In particular, the $k = 1$ scale is degenerate before clipping, so those cells are interpreted as stress tests rather than ordinary LOIO calibration.

The normalized source score views are then selected without inspecting target labels or target scores. Matched-condition mode selects the source view with the declared target condition and therefore requires condition metadata. Clean-source mode always selects clean source normals. Condition-agnostic mode takes, for each base source image, the median normalized score over all available condition views so that repeated corruptions are not counted as independent observations. Deterministic mismatched-condition routing selects a different condition solely as a negative control. Thus normalization precedes source-view selection or aggregation in the implemented protocol. Only the resulting scalar score archive and category identities are needed downstream; source patch features do not enter certification.

For a fixed target and support seed, the eligible source categories are split before candidate construction into disjoint reference, proposal, and certification sets $\mathcal{R}$, $\mathcal{P}$, and $\mathcal{C}$. The primary allocation is 0.50/0.25/0.25, with deterministic rounding that keeps every partition nonempty. The split is generated from the target-category identity and seed, is held fixed across conditions and k, and never depends on observed scores, target outcomes, or certification losses. All score views from one category remain in the same partition.

3.6. Nested CRESS Threshold Selection

Let $\mathcal{Z}_{\mathcal{R}}$ be the pooled normalized scores from the selected reference-category views. The reference categories define the source-reference rank map

$$p_{\mathcal{R}}(x) = \frac{1 + \sum_{r\in\mathcal{Z}_{\mathcal{R}}} \mathbf{1}\{r \geq \widetilde{s}_c(x)\}}{1 + |\mathcal{Z}_{\mathcal{R}}|}. \tag{6}$$

We use the term *source-reference map* deliberately: Eq. (6) is an empirical rank transformation on the source archive, and cross-category conformal validity is not presumed. Its finer grid arises from the larger reference-score pool rather than from the k target supports.

Applying the reference map to the proposal categories produces the proposal p-values $\{p_{\mathcal{R}}(x) : x \in \mathcal{P}\}$. Their distinct values generate an ordered set $\mathcal{T}$ of at most M positive candidate thresholds. If more than M distinct values occur, a deterministic evenly spaced subset of their ordered indices is retained. Certification data never influence this candidate set. The same reference map is then applied to every certification image. Given a candidate $\tau \in \mathcal{T}$, its image-unit loss is

$$L_x(\tau) = \mathbf{1}\{p_{\mathcal{R}}(x) \leq \tau\}. \tag{7}$$

All certification images are declared normal, so an alarm is a false alarm. For a certification category c, the category-unit loss is

$$L_c(\tau) = \frac{1}{n_c}\sum_{x\in c} L_x(\tau) \in [0, 1]. \tag{8}$$

The image unit targets the declared source-image mixture under iid image sampling; the category unit targets the mean loss for a new draw from a declared source-category meta-population. They are different estimands and are not interchangeable.

Let A be the number of reported nominal levels and let $n = |C|$ be the number of certification categories. For each candidate threshold, define the empirical mean category loss

$$\overline{L}_C(\tau) = \frac{1}{n}\sum_{c\in C} L_c(\tau). \tag{9}$$

The category analysis then uses the family-adjusted Hoeffding category-risk UCB

$$U_{\mathrm{H}}(\tau) = \min\left\{ 1,\ \overline{L}_C(\tau) + \sqrt{\frac{\log(2AM/\delta)}{2n}} \right\}. \tag{10}$$

For iid Bernoulli alarms from the declared source-image mixture, the image-unit sensitivity uses the exact one-sided Clopper–Pearson upper bound with tail probability $\delta/(2AM)$. The factor $2A$ allocates the family error across both unit definitions and all reported levels; M allocates it across the finite candidate family. In the selection rule below, U denotes U_{H} for category certification and the corresponding Clopper–Pearson bound for the image-unit sensitivity. CRESS returns

$$\tau^\star = \max\big(\{\tau \in \mathcal{T} : U(\tau) \leq \alpha\} \cup \{0\}\big). \tag{11}$$

The value $\tau^\star = 0$ is a deterministic fail-closed fallback because $p_{\mathcal{R}}(x) > 0$. After threshold selection, a target image can be operated on by

$$a_{\tau^\star}(x) = \mathbf{1}\{p_{\mathcal{R}}(x) \leq \tau^\star\}, \tag{12}$$

where the target score is normalized only by the target support statistics. Target test images and labels enter none of the reference, proposal, certification, or threshold-selection stages.

3.7. Certificate Scope, Feasibility, and Transfer Boundary

Proposition 1 (conditional source certificate). Conditional on the reference and proposal stages, suppose certification units are independent draws from the declared source-unit population: Bernoulli losses for the image analysis, or bounded losses in $[0, 1]$ for the category analysis. Then, with probability at least $1 - \delta$, simultaneously across the A levels and both unit definitions, the corresponding source risk of every selected threshold is at most its α.

Proof. For a fixed level, unit, and candidate, the one-sided Clopper–Pearson construction or Hoeffding's inequality fails with probability at most $\delta/(2AM)$. A union bound over M candidates, A levels, and two units gives total failure probability at most δ; independence between these bounds is unnecessary. Selection occurs only on this simultaneous event, and the zero fallback has zero alarm risk. □

Proposition 2 (Hoeffding feasibility). Under the category-level Hoeffding rule, a positive candidate can pass at level α only if

$$n \geq \frac{\log(2AM/\delta)}{2\alpha^2}. \tag{13}$$

Proof. Because $\overline{L}_C(\tau) \geq 0$, the unclipped upper bound is at least $\sqrt{\log(2AM/\delta)/(2n)}$. Requiring it not to exceed α and rearranging proves the claim. □

Proposition 3 (distribution-free all-zero lower bound). Let $\beta \in (0, 1)$ and let $L_1, \ldots, L_n$ be iid losses from an arbitrary distribution P on $[0, 1]$, with mean μ_P. Let $U_n : [0, 1]^n \to [0, 1]$ be any deterministic upper confidence bound satisfying

$$\inf_P \Pr_{P^n}\big\{\mu_P \leq U_n(L_1, \ldots, L_n)\big\} \geq 1 - \beta.$$

Then

$$U_n(0, \ldots, 0) \geq 1 - \beta^{1/n}. \tag{14}$$

Consequently, even after observing zero loss in every certification category, $U_n \leq \alpha$ is possible only if

$$n \geq \frac{\log \beta}{\log(1 - \alpha)}.$$

Proof. Write $u = U_n(0, \ldots, 0)$ and suppose, for contradiction, that $u < 1 - \beta^{1/n}$. Choose a Bernoulli distribution with mean q such that $u < q < 1 - \beta^{1/n}$. Its all-zero sample has

Placeholder: CRESS source-side threshold certification

Source supports and held-out normal views → category normalization and predeclared view selection → disjoint $\mathcal{R}/\mathcal{P}/\mathcal{C}$ split.

$\mathcal{R}$: source-reference map $p_{\mathcal{R}}$; $\mathcal{P}$: candidate thresholds $\mathcal{T}$; $\mathcal{C}$: category losses → family-adjusted risk UCB → $\tau^{\star} = \max(\mathcal{T}_{\text{pass}} \cup \{0\})$.

Figure 3: CRESS source-side threshold certification. Each source category supplies a support-fitted scorer, support-only normalization statistics, and disjoint held-out normal score views selected by a predeclared routing rule. Source categories are partitioned into reference ($\mathcal{R}$), proposal ($\mathcal{P}$), and certification ($\mathcal{C}$) roles. The reference set defines $p_{\mathcal{R}}$, the proposal set fixes the candidate family, and the certification set produces category losses for the family-adjusted risk UCB. CRESS returns the largest passing threshold or the fail-closed value zero. This is a source-domain certificate; target-category control additionally requires the transfer condition in Corollary 1. The final artwork will replace this placeholder without changing the method.

Table 1
Minimum independent certification-category counts permitted by the all-zero design calculations at $\delta = 0.05$. The distribution-free (DF) floor is the necessary lower limit in Proposition 3, first without multiplicity ($\beta = \delta$) and then under the frozen allocation $\beta = \delta/(2AM)$ with $A = 3$. "Hoeffding" is the necessary count for the declared rule in Proposition 2. All rows assume zero observed category loss; satisfying a listed count does not by itself guarantee that CRESS selects a positive threshold.

rule / candidates	$\alpha = .20$	$\alpha = .10$	$\alpha = .05$
DF floor, no multiplicity	14	29	59
DF floor, $M = 1$	22	46	94
DF floor, $M = 5$	29	61	125
DF floor, $M = 20$	35	74	152
Hoeffding, $M = 1$	60	240	958
Hoeffding, $M = 5$	80	320	1,280
Hoeffding, $M = 20$	98	390	1,557

probability $(1-q)^n > \beta$. On that event $U_n = u < q = \mu_P$, so the bound fails with probability greater than β, contradicting uniform $(1-\beta)$ coverage. Rearranging $1 - \beta^{1/n} \leq \alpha$ gives the sample-size condition. □

Proposition 3 shows that the frozen failure is not solely an artifact of Hoeffding looseness. Its Bernoulli counterexample is elementary; the contribution here is to use that necessary condition as an explicit category-budget calculation for transferable anomaly thresholds. Even for one fixed candidate, one level, no image-unit co-reporting, and no multiplicity penalty ($\beta = \delta = 0.05$), any deterministic uniformly valid distribution-free bound based only on iid category losses needs at least 14, 29, and 59 categories at $\alpha = 0.20$, 0.10, and 0.05. These are necessary lower limits, not claims that the corresponding sample sizes suffice for CRESS. Under the frozen allocation $\beta = \delta/(2AM)$ with $A = 3$ and the most favorable $M = 1$, the corresponding lower limits are 22, 46, and 94; the declared Hoeffding rule requires 60, 240, and 958.

Table 1 is a design calculation rather than an empirical result. Proposition 3 is limited to deterministic, uniformly valid, distribution-free upper bounds based solely on iid bounded category losses. Parametric or hierarchical assumptions, informative side data, and randomized confidence procedures require separate validity arguments. Adding independent source-image draws may refine the selected-mixture image analysis, but additional images or repeated support seeds cannot increase the number of independent certification categories.

Corollary 1 (conditional target transfer). On the event in Proposition 1, the risk of a particular target category is at most α if, for every proposed threshold, its normal alarm probability is no larger than the corresponding certified source risk. Alternatively, if a target category is a fresh exchangeable draw from the certification-category meta-population, the alarm risk marginalized over that random category draw is at most α.

Proof. The first statement follows by source-risk dominance and Proposition 1. Under category exchangeability, the marginal target risk equals the source meta-population risk bounded in Proposition 1. Neither argument bounds the category-conditional risk of every realized exchangeable target. □

Source-risk dominance and category exchangeability are assumptions, not consequences of normalization, source-view selection, or the rank transformation in Eq. (6). Thus $\tau^{\star}$ carries a source-domain certificate. It becomes a guarantee for a particular target only under dominance; exchangeability supplies the weaker marginal new-category statement. Direct pooled-source conformal, which uses the complete selected source pool with $\tau = \alpha$ and no disjoint proposal or certification stage, is retained as an uncertified empirical reference.

4. Experiments

4.1. Datasets and Few-Shot Splits

We evaluate MVTec AD [36] and VisA [37]. The strict external-transfer audit additionally incorporates MPDD. MVTec AD comprises industrial object and texture categories with pixel-level defect masks. VisA contains a broader range of industrial categories and serves as the primary benchmark for full-scale corruption and calibration shift. MPDD provides six metal-part categories for an external stress test that transfers from MVTec to MPDD; within-MPDD category certification is not pursued because fewer than six non-target categories remain for a three-way source split. For each category, support sets are sampled from the normal training partition using fixed seeds. The CRESS source archive is separate: it consists of held-out known-normal evaluation images from non-target source categories (the `test/good` partition for MVTec and the corresponding normal partition for VisA and MPDD). Source normal status is therefore used, but target test labels never enter reference construction, threshold proposal, or certification. Unless otherwise specified, $k \in \{1, 2, 4, 8\}$ is employed for the clean accuracy-storage and strict nested studies, whereas the target-only corruption tables use $k \in \{4, 8\}$. The $k = 1$ strict cells use the declared patch-split support calibration fallback and are interpreted as stress tests; LOIO calibration itself requires $k \geq 2$.

4.2. Evaluation Protocols

The target-only corruption audit evaluates all test images from all 15 MVTec and 12 VisA categories at $k \in \{4, 8\}$ under five nested support seeds and four corruptions. Its primary p-values follow Eq. (4): calibration scores use LOIO fits and each test score uses the full-support fit. The fold-matched construction is reported only as a sensitivity analysis. Alarm comparisons use the predeclared tolerance 10^{-6} to absorb float32 representation error at attainable grid points.

The primary empirical groups are the target-only corruption audit and the frozen strict nested CRESS audit; the category-count feasibility curves are analytic design calculations. Supporting groups cover clean ranking and storage, PCA64/PCA128 sensitivity, direct pooled-source conformal, and attainable operating levels. The strict nested protocol uses $k \in \{1, 2, 4, 8\}$, seeds from 0 to 4, $\alpha \in \{0.05, 0.10, 0.20\}$, and clean, Gaussian-noise, blur, brightness/contrast, and JPEG conditions. Each cell contains at most 120 evaluation images, with $\rho = 0.01$, PCA64, $\delta = 0.05$, and up to 20 candidate thresholds. Raw scores are first normalized using support-only statistics; condition-specific source views are then selected or aggregated as defined in Section 3.5. For every target category and seed, the eligible source categories are deterministically split using the frozen $0.50/0.25/0.25$ reference/proposal/certification allocation. This gives 7/3/4 categories for a 14-category within-MVTec source pool, 5/3/3 for an 11-category within-VisA pool, and 7/4/4 for the 15-category MVTec transfer pool. The same category partition is reused across conditions and k. The protocol evaluates MVTec and VisA within-dataset, MVTec-to-VisA, and MVTec-to-MPDD under matched-condition, clean-source, condition-agnostic, and mismatched negative-control source selection. The operational gate requires a nonzero category-certified threshold in at least 80% of target cells, a target false-alarm rate (FAR) of at most $\alpha + 0.02$, nonzero power, at least 80% no-harm, and a positive lower bound on simultaneous power gain. Zero thresholds and failed cells are retained.

4.3. Baselines and Ablations

We compare the inherited PCA residual scorer with a controlled nearest-neighbor (NN) memory bank using the same frozen DINOv2 features and few-shot split. PCA64 and PCA128 retain 64 and 128 principal directions, respectively; PCA64 is the default reliability substrate. For the source-assisted study, direct pooled-source conformal is an uncertified baseline: it uses the complete selected source pool with $\tau = \alpha$, but performs no reference, proposal, or certification split and applies no UCB gate. The controlled memory-bank implementation is not an official reproduction of PatchCore or AnomalyDINO; it isolates scoring and storage trade-offs under a common protocol. Official AnomalyDINO code is also run on MVTec and VisA under its released protocol and reported as a separate accuracy reference. WinCLIP [11] is included only through published values because no official implementation is available. Official SubspaceAD representative runs serve as a novelty guardrail: the frozen DINOv2 subspace residual is inherited, and the contribution begins with the reliability analysis downstream of that score.

4.4. Metrics

Ranking metrics are distinguished from reliability metrics. AUROC and AP are calculated from the raw PCA residual score $s(x)$. Reliability is evaluated operationally by empirical FAR at fixed α, detection power, alarm precision, the empirical cumulative distribution function of normal p-values, and the fraction of cells receiving a positive certified threshold. Storage is quantified as retained ranker state per category. Corruption views are controlled stress tests and are not treated as independent images or as a proxy for all deployment shifts.

4.5. Reproducibility and Claim Rules

We conduct all experiments on a single NVIDIA RTX 5090 graphics processing unit (GPU) using PyTorch 2.12.1. DINOv2 patch features are cached in 32-bit floating-point (fp32) format. For reproducibility, the frozen commit, DINOv2 source and weight hashes, environment specifications, dataset counts, per-image predictions, base-image identities, support manifests, corruption parameters, source partitions, candidate-threshold bounds, and Secure Hash Algorithm 256-bit (SHA-256) checksums are documented. Automated integrity checks encompass 1,500 MVTec, 1,200 VisA, and 600 MPDD view cells, identifying duplicate cells, missing support statistics, or overlap between support and test sets.

Table 2
Target-only LOIO operating points. The floor is the smallest p-value, $1/(k+1)$; "attainable" is the largest p-value grid point not exceeding the nominal α, or zero if none exists. Within each corruption, FAR and detection are pooled over classes and seeds; the table then averages the four corruption-specific rates.

dataset	k	floor	nominal α	attainable	FAR	detection
VisA (full)	4	0.200	0.01	0.000	0.000	0.000
			0.05	0.000	0.000	0.000
			0.10	0.000	0.000	0.000
			0.20	0.200	0.142	0.595
	8	0.111	0.01	0.000	0.000	0.000
			0.05	0.000	0.000	0.000
			0.10	0.000	0.000	0.000
			0.20	0.111	0.094	0.555
MVTec (full)	4	0.200	0.01	0.000	0.000	0.000
			0.05	0.000	0.000	0.000
			0.10	0.000	0.000	0.000
			0.20	0.200	0.278	0.880
	8	0.111	0.01	0.000	0.000	0.000
			0.05	0.000	0.000	0.000
			0.10	0.000	0.000	0.000
			0.20	0.111	0.200	0.868

Table 3
Target-only LOIO FAR, anomaly-detection rate, and precision at the first reported attainable level, α=0.20. Results use five nested support seeds, are pooled over images within each dataset, corruption, and k, and retain every normal and anomalous test image. Alarms fire at $p \leq \alpha$ using the comparison tolerance described in the text.

dataset	corruption	k	FAR	detection	precision
VisA (full)	blur	4	0.144	0.606	0.825
		8	0.094	0.566	0.870
	bright/contr.	4	0.146	0.596	0.821
		8	0.098	0.557	0.864
	Gauss. noise	4	0.133	0.575	0.829
		8	0.081	0.538	0.881
	JPEG	4	0.145	0.602	0.822
		8	0.104	0.559	0.858
MVTec (full)	blur	4	0.219	0.880	0.904
		8	0.139	0.870	0.936
	bright/contr.	4	0.214	0.882	0.906
		8	0.133	0.870	0.939
	Gauss. noise	4	0.341	0.873	0.857
		8	0.281	0.860	0.878
	JPEG	4	0.336	0.883	0.860
		8	0.248	0.872	0.892

Reliability methods cannot claim ranking improvement unless the raw score is altered; image-level and category-level certificates are not interchangeable.

5. Results

5.1. Shift-Induced False-Alarm Inflation

Table 2 first separates unattainable nominal levels from observed false-alarm behavior. With $k = 4$, the smallest p-value is $1/(k + 1) = 0.2$, so nominal levels 0.01, 0.05, and 0.10 cannot raise an alarm. With $k = 8$, the floor decreases to $1/9 \approx 0.111$, but the same three levels remain below it. Their zero FAR and detection values are finite-resolution artifacts, not evidence of conservative false-alarm control.

At nominal $\alpha = 0.20$, the effective threshold is 0.20 for $k = 4$ but only $1/9$ for $k = 8$, because $1/9$ is the largest $k = 8$ grid point not exceeding 0.20. On VisA, the aggregate FAR/detection pairs are 0.142/0.595 at $k = 4$ and 0.094/0.555 at $k = 8$; on MVTec they are 0.278/0.880 and 0.200/0.868. These aggregates first pool images over categories and five nested support seeds within each corruption and then average the four corruption-specific rates. The nominal level therefore induces different effective thresholds across k, while the empirical FAR itself is not restricted to the p-value grid.

Table 3 resolves the aggregate by corruption. The headline 0.341 is the empirical FAR on Gaussian-corrupted MVTec at $k = 4$, not the MVTec average or a population FAR. It is the largest dataset-level aggregate in the frozen grid and is approximately 1.7 times the nominal 0.20. Category-level FARs are heterogeneous: after pooling within category over the five seeds, the median is 0.215, the interquartile range is 0.169 to 0.481, and 9 of 15 categories exceed 0.20. Thus, the aggregate is neither universal across categories nor attributable to only one category. JPEG is similarly anti-conservative at 0.336. At $k = 8$, blur and brightness/contrast are conservative at 0.139 and 0.133, whereas Gaussian noise and JPEG remain above nominal at 0.281 and 0.248. Increasing k mitigates but does not eliminate the mismatch.

VisA behaves differently: FAR remains between 0.133 and 0.146 at $k = 4$ and between 0.081 and 0.104 at $k = 8$. The shift effect is therefore dataset- and condition-dependent, not a universal inflation result. Detection remains nonzero at every displayed operating point; precision is descriptive because it depends on benchmark anomaly prevalence. An fp32 representation of $1/5$ may equal 0.20000000298, so exact comparison with 0.2 can inadvertently exclude floor-level alarms. All reported thresholding uses the fixed numerical tolerance declared in the protocol.

The normal-p-value audit explains the direction of these deviations. Conformal p-values satisfy $P(p \le \alpha) \le \alpha$ for normal data only under the relevant exchangeability conditions. Because a k-score p-value lies on the grid $\{j/(k+1)\}$, a continuous Kolmogorov–Smirnov test against $U(0,1)$ is inappropriate. Figure 4 instead compares the empirical normal-p-value cumulative distribution function (CDF) with the ideal discrete-uniform reference at each attainable point. VisA at $k = 4$ is conservative, with a largest displayed negative gap of -0.087 at 0.6; at $k = 8$, its first grid point remains conservative although small anti-conservative gaps appear at higher grid points. MVTec is anti-conservative at both k values, especially under Gaussian noise and JPEG. For Gaussian-corrupted MVTec at $k = 4$, $\widehat{F}(0.2) = 0.341$ versus the idealized reference value 0.20, and the gap grows to 0.204 at the 0.4 grid point.

Rows share calibration sets, categories, and repeated test images across seeds. We therefore treat the CDF comparison as descriptive and do not attach an iid goodness-of-fit test to the pooled points. The results show that corrupted normal scores increase relative to clean support calibration scores, but do not identify a causal mechanism. On full MVTec at $k = 4$, the fold-matched LOIO sensitivity increases clean FAR from 0.212 to 0.257 and power from 0.880 to 0.906. Removing the fold-fit asymmetry therefore does not restore false-alarm control and incurs a less favorable FAR/power trade-off in this audit.

5.2. Independent-Category Feasibility

Figure 1 and Table 1 are analytic design calculations, not fitted learning curves. In the optimistic multiplicity-free case, the all-zero distribution-free floor crosses $\alpha = 0.20$, 0.10, and 0.05 only at 14, 29, and 59 independent categories. These counts are necessary within the scope of Proposition 3; they are not sufficient sample sizes for a positive CRESS threshold. The frozen family allocation with $A = 3$ and favorable $M = 1$ increases the lower limits to 22, 46, and 94, while the corresponding Hoeffding rule requires 60, 240, and 958.

Candidate multiplicity widens the gap. At $M = 20$, the family-adjusted distribution-free lower limits become 35, 74, and 152, and the Hoeffding requirements become 98, 390, and 1,557. The three or four certification categories available in the frozen audit lie below even the most optimistic 14-category requirement at $\alpha = 0.20$. This comparison answers a feasibility question before any target performance is inspected: a nonzero distribution-free certificate of marginal new-category risk is unavailable at the present category budget, even under zero observed category loss. The result does not rule out parametric, hierarchical, randomized, or side-information procedures; such methods require separate assumptions and validity arguments.

5.3. CRESS Boundary Test: Categories versus Images

Table 4 tests whether the analytic shortage binds in the frozen CRESS protocol. The 960 gate configurations comprise four source-to-target jobs, four source-view modes, four k values, three α levels, and five conditions. They are configurations in a frozen grid, not 960 independent statistical trials. Every category-certified target cell receives $\tau^\star = 0$, and no configuration reaches the required 80% nonzero-threshold rate. The smallest candidate UCB is 0.950 on MVTec, 1.000 on VisA, 0.961 for MVTec-to-VisA, and 0.986 for MVTec-to-MPDD. Even the smallest value is 4.75 times the largest tested $\alpha = 0.20$.

The fail-closed result agrees with the design calculation and is not a compute or optimization failure. Proposition 2 requires at least 60 categories for the favorable $M = 1$ Hoeffding gate at $\alpha = 0.20$, while Proposition 3 requires at least 14 categories even for the optimistic multiplicity-free distribution-free case and at least 22 under the favorable frozen allocation. Only three or four certification categories are available. Because the operational gate is conjunctive, the zero nonzero-threshold rate already determines failure; downstream target-performance checks cannot convert $\tau^\star = 0$ into a usable alarm threshold. The strict experiment therefore confirms that the predicted category-count boundary binds throughout the declared grid; it does not provide 960 replications of the theoretical result.

The image-unit block changes both the sample size and the estimand. Under its conditional iid source-image assumption, 72 to 191 selected source-image units per cell reduce the smallest UCB to 0.039 to 0.048. Positive thresholds occur in 36.7% of MVTec, 60.3% of VisA, 41.5%

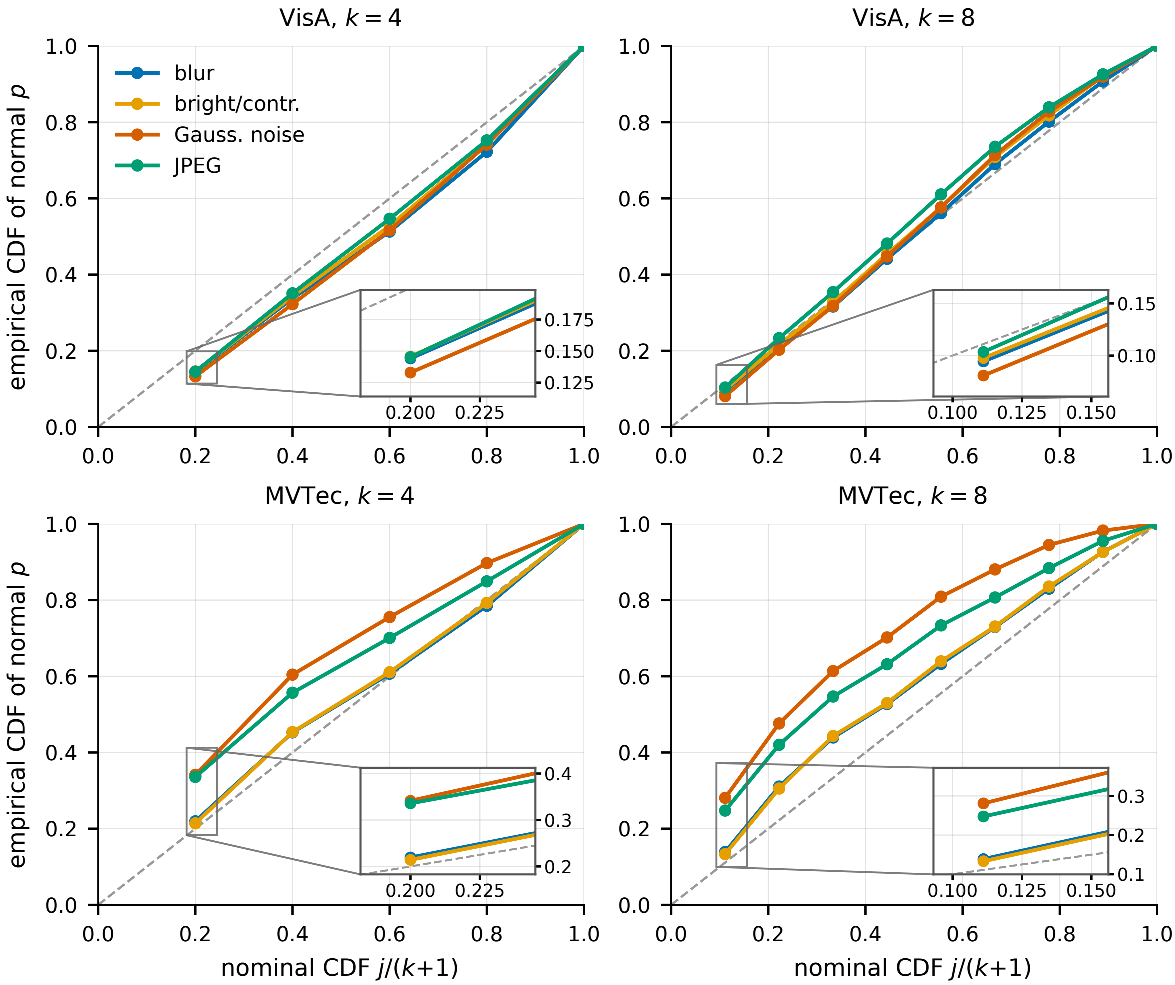

Figure 4: Empirical CDF of normal-image LOIO p-values at each attainable point $j/(k+1)$ versus the idealized exchangeable reference (dashed diagonal), using five nested support seeds. Points above the diagonal indicate excess false alarms relative to the reference; lower-right insets and connector rays enlarge the first attainable grid region.

Table 4
Frozen strict nested CRESS audit across four source-to-target jobs, four source-view modes, $k \in \{1, 2, 4, 8\}$, three α levels, and five conditions (960 configurations), each evaluated over five support seeds. "Min UCB" is the smallest observed candidate upper bound. Nonzero is the fraction of target cells receiving $\tau > 0$. Category and image rows target different estimands; image rows additionally assume iid source-image units.

source → target	certification unit	units/cell	min UCB	nonzero	interpretation
MVTec → MVTec	category	4	0.950	0.000	marginal category-risk gate fails
VisA → VisA	category	3	1.000	0.000	marginal category-risk gate fails
MVTec → VisA	category	4	0.961	0.000	marginal category-risk gate fails
MVTec → MPDD	category	4	0.986	0.000	marginal category-risk gate fails
MVTec → MVTec	source image	75 to 191	0.039	0.367	iid-image sensitivity
VisA → VisA	source image	150 to 180	0.042	0.603	iid-image sensitivity
MVTec → VisA	source image	72 to 159	0.048	0.415	iid-image sensitivity
MVTec → MPDD	source image	81 to 190	0.040	0.370	iid-image sensitivity

of MVTec-to-VisA, and 37.0% of MVTec-to-MPDD target cells. These rates remain below the 80% operational gate overall. More importantly, their risk refers to image draws from the selected source mixture. The additional images do not create independent categories and cannot be relabeled as evidence for a marginal new-category certificate. Thus, the same source data can be informative for the image-mixture estimand and insufficient for a distribution-free category-transfer estimand.

Table 5
Direct pooled-source conformal under matched-condition source selection. FAR and power are means over target class, $k \in \{1, 2, 4, 8\}$, seeds from 0 to 4, and five conditions. The baseline uses the complete selected source pool with $\tau = \alpha$ and has no reference, proposal, or certification split.

source → target	α	FAR	power
MVTec → MVTec	0.05	0.075	0.404
	0.10	0.131	0.566
	0.20	0.242	0.754
VisA → VisA	0.05	0.060	0.320
	0.10	0.111	0.466
	0.20	0.208	0.639
MVTec → VisA	0.05	0.028	0.171
	0.10	0.048	0.282
	0.20	0.099	0.441
MVTec → MPDD	0.05	0.030	0.186
	0.10	0.066	0.283
	0.20	0.157	0.423

Table 5 reports direct pooled-source conformal as an uncertified empirical reference. It uses the complete selected source pool with $\tau = \alpha$ and no disjoint reference, proposal, or certification categories. Under matched-condition source selection, it has nonzero power in all four source-to-target jobs, but mean FAR exceeds nominal within both MVTec and VisA. Transfer is more conservative and less powerful. Direct pooling supplies finer operating points than target-only LOIO, but characterizes empirical risk under the selected source mixture rather than certified new-category risk.

5.4. Supporting Ranking Substrate and Storage

Table 6 verifies that the reliability conclusions are studied on a viable, compact ranking substrate. The official AnomalyDINO reproduction achieves the highest image AUROC at each reported k on both datasets, but follows its released evaluation protocol and therefore serves as an external accuracy reference rather than evidence of superiority under our controlled protocol. WinCLIP is likewise a cited reference: its paper reports AUROC and area under the precision-recall curve (AUPR) at one and four shots, but not an average-precision implementation verified to match ours. We mark its AP entries N/C instead of mixing the reported AUPR values into the controlled AP comparison. WinCLIP also does not report category-specific storage under our accounting or an eight-shot result.

Compared with the controlled DINOv2 nearest-neighbor ranker, PCA64 yields lower AUROC by 0.010, 0.005, and 0.003 on MVTec and higher AUROC by 0.019, 0.008, and 0.007 on VisA at $k = 1, 4, 8$, respectively. PCA64 is a compact supporting ranker, not a ranking contribution or state-of-the-art result. On VisA, PCA128 increases class-averaged AUROC over PCA64 by 0.011, 0.015, and 0.017. Boldface is restricted to methods evaluated under our common protocol; external AnomalyDINO and WinCLIP references are excluded.

Storage and compute accounting. Storage excludes the shared frozen DINOv2 ViT-S/14 backbone, which is stored once for all categories. With $d = 384$ fp32 features, the PCA mean and basis require $(m+1)d$ scalars: 0.095 MiB for PCA64 and 0.189 MiB for PCA128. Support-normalization scalars are negligible at the displayed precision and no auxiliary calibration head is included. The controlled memory bank stores 2.005 MiB at $k = 1$ and is capped at 4096 vectors, or 6.000 MiB, for larger k. With cached features, PCA-residual scoring takes approximately 1.3 ms per image, compared with 5 to 13 ms for controlled nearest-neighbor scoring. These timings exclude shared backbone inference. CRESS subsequently retains normalized scalar reference scores and partition metadata rather than source patch features.

The clean benchmark uses max pooling, whereas the conformal pipeline uses the top-1% mean as a smoother operational score. Table 6 is therefore evidence that the inherited substrate is compact and viable under its declared clean protocol; it is not used to claim that aggregation is immaterial or to attribute any reliability difference to the PCA ranker.

6. Limitations

The feasibility result has a deliberately narrow statistical scope. Proposition 3 applies to deterministic, uniformly valid, distribution-free upper confidence bounds based only on iid bounded category losses. It does not rule out nonvacuous certification from fewer categories under pre-specified parametric or hierarchical models, informative side data, or randomized procedures. Such approaches exchange distribution-free generality for additional assumptions and require their own coverage analyses. The counts 14, 29, and 59 are optimistic necessary limits without multiplicity, not sufficient sample sizes for CRESS. The frozen family allocation and declared Hoeffding rule require more.

The category interpretation also depends on the unit definition. The strict audit contains only three or four certification categories, and repeated images, corruption views, or support seeds do not constitute additional independent category draws. Image-unit certificates under the selected-mixture iid model therefore address a different estimand rather than a partial certificate for a new category. CRESS certifies source-domain risk under the assumptions of Proposition 1. The benchmark categories are a fixed heterogeneous collection, so their interpretation as independent draws from a source-category meta-population is a modeling assumption rather than an empirically testable fact in these data. Applying the selected threshold to a particular target additionally requires source-risk dominance. Category exchangeability supports only risk marginalized over a random target-category draw; it does not control every realized category. Support-only normalization and source-view selection establish neither condition. Matched source selection also assumes reliable condition metadata.

Table 6
Clean image-level ranking performance and category-specific ranker state in mebibytes (MiB). Controlled DINOv2 NN and PCA64/PCA128 share our feature, split, metric, and storage-accounting protocol and are averaged over classes and five seeds. The official AnomalyDINO reproduction uses three seeds under its released protocol; WinCLIP values are reported in [11]. Bold marks the best value only among common-protocol rows in each dataset/k block. "–" denotes not evaluated, N/C not directly comparable, and N/R not reported.

Dataset	k	Method	AUROC ↑	AP ↑	MiB ↓
MVTec	1	Official AnomalyDINO	0.965	0.982	–
		WinCLIP (reported)	0.931	N/C	N/R
		Controlled DINOv2 NN	**0.914**	**0.952**	2.005
		PCA64 (controlled)	0.904	0.951	**0.095**
	4	Official AnomalyDINO	0.976	0.984	–
		WinCLIP (reported)	0.952	N/C	N/R
		Controlled DINOv2 NN	**0.942**	**0.967**	6.000
		PCA64 (controlled)	0.937	**0.967**	**0.095**
	8	Official AnomalyDINO	0.980	0.990	–
		Controlled DINOv2 NN	**0.948**	0.970	6.000
		PCA64 (controlled)	0.945	**0.972**	**0.095**
VisA	1	Official AnomalyDINO	0.857	0.866	–
		WinCLIP (reported)	0.838	N/C	N/R
		Controlled DINOv2 NN	0.804	0.809	2.005
		PCA64 (controlled)	0.823	0.834	**0.095**
		PCA128 (controlled)	**0.834**	**0.842**	0.189
	4	Official AnomalyDINO	0.912	0.918	–
		WinCLIP (reported)	0.873	N/C	N/R
		Controlled DINOv2 NN	0.862	0.861	6.000
		PCA64 (controlled)	0.870	0.883	**0.095**
		PCA128 (controlled)	**0.885**	**0.895**	0.189
	8	Official AnomalyDINO	0.926	0.929	–
		Controlled DINOv2 NN	0.873	0.870	6.000
		PCA64 (controlled)	0.880	0.891	**0.095**
		PCA128 (controlled)	**0.897**	**0.905**	0.189

The empirical evidence is limited to MVTec, VisA, and MPDD. The controlled shift study covers Gaussian noise, blur, brightness/contrast, and JPEG corruption, not the full range of temporal drift, sensor replacement, process changes, correlated failures, or deployment prevalence. The empirical FAR of 0.341 is the largest dataset-level aggregate in this frozen synthetic-corruption grid, not a population failure probability or a universal response to Gaussian noise. The image-unit sensitivity additionally assumes independent source-image draws; repeated or correlated production images would reduce its effective evidence. Prospective factory evidence is needed to determine whether the same failure mode and category-count assumptions hold under real operational shifts.

Finally, the frozen DINOv2 PCA residual is an inherited ranking substrate. Official AnomalyDINO and reported WinCLIP results follow different protocols and serve as external references; the paper does not claim ranking state of the art. LOIO calibration and full-support test residuals remain asymmetric, and the pooled discrete-grid audit diagnoses rather than restores exchangeability. Reported scoring latency uses cached DINOv2 features and excludes backbone inference, preprocessing, and input/output overhead.

7. Conclusion and Future Work

This study has established that the reliability of few-shot anomaly thresholds depends on both the amount and the statistical unit of normal evidence. At nominal $\alpha = 0.20$, target-only LOIO has reached an empirical FAR of 0.341 on Gaussian-corrupted MVTec at $k = 4$, showing that an attainable operating point need not remain stable under shift. The finite floor $1/(k+1)$ has explained when a target-only rule is structurally silent, but crossing that floor has not supplied an exchangeability guarantee.

The category-count analysis has quantified the separate obstacle faced by source-assisted certification. Even after all-zero observed category losses, optimistic multiplicity-free distribution-free 95% certification has required at least 14, 29, and 59 independent categories at $\alpha = 0.20, 0.10$, and 0.05. The frozen allocation and Hoeffding rule have required more, whereas the strict audit has provided only three or four certification categories. Every category-certified target cell has consequently received the fail-closed threshold $\tau^\star = 0$ across all 960 frozen configurations, whose shared boundary condition precludes interpreting them as independent trials. In contrast, image-unit analyses under the selected-mixture iid model have selected positive thresholds in 36.7% to 60.3% of target cells, but have targeted the selected source mixture rather than marginal risk over a new-category draw.

The practical contribution has therefore been a study-design discipline: distinguish ranking from threshold reliability, report attainable rather than nominal operating points, audit normal false alarms under declared shifts, define the certification unit before observing results, and count independent categories when the claim concerns a new category. Future work should collect category-rich source archives or introduce explicit parametric, hierarchical, or side-information assumptions with independently verified validity. It should also test real production shifts, temporal dependence, condition-estimation error, and end-to-end deployment cost. These directions can improve usefulness, but they should not relabel repeated images as category evidence, confuse marginal with category-conditional risk, or convert a source-domain certificate into unconditional target control.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets analyzed during the current study are publicly available from the MVTec AD dataset page, the VisA repository, and the MPDD repository. Derived per-image predictions, evaluation summaries, sampling and source-partition manifests, and checksums are available in the project repository.

Code availability

The experimental code, corruption pipeline, integrity audits, and table and figure generators are available in the project repository. The primary GPU run uses frozen scientific revision `e7f175990b02`; its complete result archive is recorded at revision `74fdfd00f348`.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used OpenAI Codex to support code review, software-test generation, statistical-claim auditing, and language editing. No experimental observation was generated or altered by the tool. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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